Arash Ahmadi

arash.ahmadi@ou.edu | (+1) 405 361 7368| arash-ahmadi.com | Norman, OK, United States

Education

- **Doctor of Philosophy (Ph.D.), Electrical and Computer Engineering** Aug 2024 – Current
  University of Oklahoma, Norman, OK, USA
  Research focus: Fine-tuning small language models for efficient deployment on resource-constrained edge devices, with applications in agentic systems and diverse domains.

- **Bachelor of Computer Engineering, Software Engineering** Sep 2018 – Feb 2023
  University of Kurdistan, Sanandaj, Kurdistan, Iran
  GPA: 3.88/4.0
  Thesis: *Efficient brute-force state space search for Yin-Yang puzzle*

Publications

Journal Articles

[J1] **A. Ahmadi**, S. S. Sharif, and Y. M. Banad, "A comparative study of sampling methods with cross-validation in the fedhome framework," *IEEE Transactions on Parallel and Distributed Systems*, 2025.

[J2] **A. Ahmadi** and A. Khorramian, "Efficient Brute-force state space search for Yin-Yang puzzle," *The Journal of Supercomputing*, vol. 80, no. 3, pp. 3066-3088, 2024.

Conference Proceedings

[C1] **A. Ahmadi**, S. Sharif, and Y. M. Banad, "LLM-Powered HFACS Analysis of ASRS Narratives via MCP Bridge for Enhanced Aviation Safety Debriefs," in the *2026 AIAA SciTech Forum*, Orlando, FL, Jan 2026.

[C2] **A. Ahmadi**, S. Sharif, and Y. M. Banad, "Towards Transparent Artificial Intelligence: Exploring Modern Approaches and Future Directions," in *2024 Conference on AI, Science, Engineering, and Technology (AIxSET)*, pp. 248-251, IEEE, 2024.

Preprints

[P1] **A. Ahmadi**, S. Sharif, and Y. Banad, "Improving Aviation Safety Analysis: Automated HFACS Classification Using Reinforcement Learning with Group Relative Policy Optimization," *arXiv preprint arXiv:2508.21201*, 2025.

[P2] **A. Ahmadi**, S. Sharif, and Y. M. Banad, "MCP Bridge: A Lightweight, LLM-Agnostic RESTful Proxy for Model Context Protocol Servers," *arXiv preprint arXiv:2504.08999*, 2025.

Research Experience

- **Graduate Research Assistant**, Inquire Lab Aug 2024 – Present
  University of Oklahoma, Norman, OK, USA (Supervisors: Dr. Yaser Mike Banad, Dr. Sarah Sharif)
  - Focusing on optimizing and fine-tuning language models for deployment on resource-constrained edge devices, exploring their applications in agentic systems and diverse domains such as aviation safety and health monitoring.

- **Research Assistant** Feb 2023 – Aug 2024
  University of Kurdistan (Supervisor: Dr. Amanj Khorramian)
  - Conducted research on theoretical computer science projects in the fields of Graphs, Puzzles, and Parallel Algorithms.
  - Proposed, implemented, and parallelized a novel algorithm for the Yin-Yang puzzle; this work included analyzing the results and authoring the majority of the publication.

Awards and Honors

- William H. Barkow Graduate Fellowship, University of Oklahoma 2025-2026
- William H. Barkow Scholarship, University of Oklahoma 2025-2026
- Journal Paper Award, Electrical and Computer Engineering, University of Oklahoma 2025
  For the publication: "A comparative study of sampling methods with cross-validation in the fedhome framework"

Teaching and Mentoring

- **Instructor and Lead Developer** 2025
  Cybersecurity Essentials Workshop, University of Oklahoma
  - Led the technical development for a five-day intensive workshop for IT professionals, sponsored by the Oklahoma Office of Homeland Security and FEMA.
  - Authored and presented modules on emerging cyber threats, ransomware prevention, phishing analysis, secure network design, and firewall best practices.
  - Designed and developed the official workshop website and a custom content management platform.
- **Teaching Assistant** Fall 2019 – Spring 2020
  University of Kurdistan
  - Courses: Advanced Programming with Java (Spring 2020), Fundamentals of Programming in C (Fall 2019).
  - Conducted weekly review sessions and provided direct feedback on projects and assignments.

Selected Projects

- **Inquire Lab Research Website (inquirelab.ai):** Designed and developed the official website for the Inquire Lab with Next.JS. The site serves as a public-facing platform to showcase research projects, publications, and team members.
- **Task Scheduling Optimization (Python):** Developed an application with a UI to implement and compare heuristic search algorithms (e.g., Greedy, Hill-climbing, Genetic Algorithm) for task scheduling.
- **Traffic Prediction in Edge Environments (Python):** Implemented a federated learning approach with a GRU network in TensorFlow to predict traffic. This method enhanced privacy and reduced communication costs.
- **Video Synchronization Application (Flutter):** Built an app to synchronize video playback across Android devices by integrating `flutter_vlc_player` and ZeroTier/Tailscale.
- **Othello Game with AI (Java):** Implemented Othello (Reversi) with a greedy algorithm-based AI opponent, save/load functionality, and move highlighting.

Leadership and Activities

- **Engineering Officer**, Game Developer's Association, University of Oklahoma Aug 2024 – Dec 2024
  - Delivered technical presentations on engineering principles in video game development.
  - Developed an interactive Ouija board powered by the Gemini 2.0 Flash large language model for a university-wide Halloween escape room event.

References

Yaser (Mike) Banad, PhD
Associate Professor
School of Electrical and Computer Engineering
The University of Oklahoma
bana@ou.edu

Sarah Sharif, PhD
Assistant Professor
School of Electrical and Computer Engineering
The University of Oklahoma
s.sh@ou.edu